

Research paper

Transfer learning-based updating method of transient stability assessment model

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ABSTRACT

An updating method for data-driven based transient stability assessment model of power systems is proposed in this paper. To cope with potential faults in the real-time operation of power systems, the parameters of models can be modified online through the proposed method. Firstly, according to the long-short term memory (LSTM) network, transient stability assessment models are initially trained offline by expected faults. Then, the feature distribution differences between expected faults and potential faults are evaluated by the time-sequence maximum mean discrepancy (TMMD). Compared with the traditional maximum mean discrepancy (MMD), the proposed TMMD approach takes the temporal properties of transient stability into account, which can reflect the relationship between fault samples and time series better. Finally, the parameters of models are adjusted by transfer learning to reduce the feature distribution differences, by which the updated models can be more suitable for the different but related potential faults. Therefore, through the proposed model updating method, the extensibility of evaluation models is greatly enhanced, which is more in line with the practical conditions of power systems. In this paper, the effectiveness of the proposed method has been demonstrated by the evaluation results in the IEEE 39-bus system and a realistic power system.

1. Introduction

With the development of AC-DC hybrid power grids and the increasing application of electronic devices, the dynamic characteristics of power systems become more and more complex (Hartmann et al., 2019; Krishan and Suhag, 2019; Chen et al., 2023; Chen and Bu, 2024; Zhu et al., 2021). This phenomenon inevitably aggravates the difficulty of transient stability assessment (TSA). Traditional TSA methods include the time domain method (Zadkhast et al., 2015; Tang et al., 1994) and the direct method (Zou et al., 2003; Shuai et al., 2019; Guo et al., 2020). Owing to the detailed modeling process, evaluation results of the time domain method can achieve high precision. However, it cannot meet the requirements of online assessment with large computation (Yan et al., 2019). As for the direct method with clear physical significance, TSA results can be quantitatively analyzed. However, the demand of evaluation accuracy cannot be satisfied for the simplified model (Liu et al., 2018).

In recent years, with the development of the wide area measurement system (WAMS) and artificial intelligence, machine learning-based methods have been successfully applied to the field of TSA (Wang and

Ouyang, 2022; Wang et al., 2023; Chen et al., 2023). By figuring out the mapping relationship between data and transient stability, the rapid and precise evaluation results can be acquired by these methods. The machine learning-based approaches contain two parts: shallow learning and deep learning.

Shallow learning methods have been widely applied in TSA, including random forest (RF) (Suprême et al., 2018), random vector functional link (RVFL) (Pan et al., 2020), artificial neural network (ANN) (Ostad-Ali-Askari and Shayannejad, 2021; Ostad-Ali-Askari and Shayan, 2021), decision tree (DT) (Aghamohammadi and Abedi, 2018; Guo and Milanović, 2014), extreme learning machine (ELM) (Chen et al., 2016; Li and Yang, 2017), support vector machine (SVM) (Wang et al., 2016) and so on. Although the methods mentioned above can achieve rapid evaluation effects, the prediction precision is limited by their simple structure.

Acting as another branch of machine learning-based methods, deep learning has been gradually developed in TSA (Wang and Wu, 2022; Chen et al., 2022; Chen and Wang, 2021; Pan et al., 2018; Gu et al., 2023). Compared with the algorithms of shallow learning, deep learning methods have better capabilities of data mining and nonlinear fitting.

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Thus, TSA rules can be better learned by deep learning models. In literature (Mahdi and Genc, 2018), a method based on stacked sparse auto-encoder (SSAE) is presented to extract the important features and reduce the dimension of measurements from the whole fault-on period. In literature (Yan et al., 2019), without the losses of accuracy, a cascaded convolutional neural network (CNN) is designed to terminate time-domain simulation as soon as possible to reduce the average simulation time for the batch evaluation. In literature (Du et al., 2021), a TSA method based on bidirectional long short-term memory (BiLSTM) network is proposed to improve the reliability of critical sample prediction results.

When the training set of the TSA model takes various kinds of expected faults into account, the trained model can usually achieve satisfied evaluation results. However, expected faults cannot cover all the fault cases that occur in power systems. Therefore, there is a deviation between the expected faults and the actual situation. The trained model may not be adaptive to some unexpected potential faults in the TSA process.

To deal with the problem mentioned above, transfer learning is employed for online updating, which impels the updated model closer to reality. By transfer learning methods, the model learned from expected faults can be transferred to potential faults by taking the advantage of similarities between data, tasks, or models. At present, the main work of transfer learning contains three aspects (Pan and Yang, 2010). Firstly, according to similarities between training data and test data, the certain parts of training data can be reused for learning in test data by reweighting (Tian et al., 2020; Ren et al., 2020). However, parts of the training data with low weights may lose some important information of samples. Secondly, it is assumed that some parameters or prior distributions of hyperparameters in models can be shared in the tasks of training and testing. The transferred knowledge will be encoded into the shared parameters or priors (Gao et al., 2008). Thus, the generalization ability of trained models is improved by fine-tuning the shared parameters. In literature (Kim et al., 2023), a deep transfer learning method using ImageNet pre-trained deep convolutional neural network for TSA is proposed. In literature (Li et al., 2023), a model structure is crafted that integrates model transfer learning and Transformer. The proposed method not only expands the model fault assessment scenario, but also improves the model accuracy. However, due to the adjustment of partially shared parameters or prior distributions, the improvement effects of migration are limited. Thirdly, the main point is to find out a feature representation that can reduce the differences of feature distributions between the training data and test data. In general, the performance of feature representation relies on the digging ability of machine learning methods. In literature (Ren and Xu, 2020), based on the joint distribution adaptation (JDA), the proper feature representation can be obtained by reducing the marginal distribution differences and the conditional distribution differences. In Literature (Ren et al., 2024), a reliable transfer learning method is proposed for pre-fault dynamic safety assessment of power system, which can improve the evaluation performance of unknown faults. However, the proper feature representation is explored before model training, which will ruin the feature information from the original samples. In short, an efficient approach based on transfer learning is demanded to realize the dynamic updating in TSA, so that the unexpected faults can also be evaluated accurately through the updated model.

The main contribution of this paper is the proposal of an updating methodology combining the feature transfer and model transfer for TSA. On the one hand, based on the conception of feature transfer, by adding a domain adaptive layer to the model, the differences of feature distributions between expected faults and potential faults can be evaluated

accurately by TMMD. Compared with the feature transfer alone, the method proposed in this paper can better explore the complex nonlinear mapping relationship between the features and transient stability with the consideration of temporal properties of transient stability. On the other hand, based on the conception of model transfer, the distribution differences of samples are attached to the total loss function to update TSA models. Compared with the individual method of model transfer, the global model parameters can be adjusted to achieve a better update effect.

2. Proposed methodology

2.1. Application scenario

The fault samples which are expected to occur or have occurred for model training are treated as the expected faults. By applying the database of expected faults, the mapping relationship between the sample features and system stability can be obtained by the model training. The trained model is able to judge the transient stability of expected faults well. However, in real operation, some fault cases possess different feature distributions from the expected faults, which are regarded as the potential faults. Obviously, the previous mapping function trained by the expected faults is not suitable for these potential faults. Therefore, an updated method is proposed to reduce the feature distribution differences between expected faults and potential faults, so that the updated model can also be adapted to the potential faults. The evaluation criterion transferring to potential faults is presented in Fig. 1.

As demonstrated in Fig. 1, the sample set composed of expected faults is adopted to train the TSA model, which generates the classification boundary to distinguish stable samples from unstable samples. Then, after appropriate extension, the classification boundary is directly applied to the potential faults. Before transfer learning, because of the different probability distributions of features, the performance of classification boundary is not satisfied with the large misjudgements of potential faults. Through transfer learning, the feature distributions of samples under different fault conditions will tend to be consistent, which can improve the evaluation accuracy of potential faults.

2.2. Integral conception

The focal point of the proposed approach is to evaluate and then reduce the differences of feature distributions between the expected faults and potential faults.

In TSA, although the data distributions between the expected faults and potential faults are different, their categories and tasks are consistent. Therefore, as long as the differences of feature distributions are reduced, the model can perform well. In view of the points above, maximum mean discrepancy (MMD) is used to evaluate the differences of feature distributions (Gretton). On the basis of MMD, considering that the sample features are highly correlated in time series, a time-sequence maximum mean discrepancy (TMMD) is proposed in this paper to better evaluate the differences of expected faults and potential faults in the model updating.

In this paper, two aspects of work are carried out to reduce the differences of feature distributions. On the one hand, the capability of feature extraction will affect the quality of transfer learning. Considering that the features of fault samples possess obvious temporal properties, the information of features is extracted by LSTM, which can obtain the mapping relationship between the sample data and transient stability in time series well. On the other hand, the differences of feature distributions evaluated by TMMD are added to total loss function as a

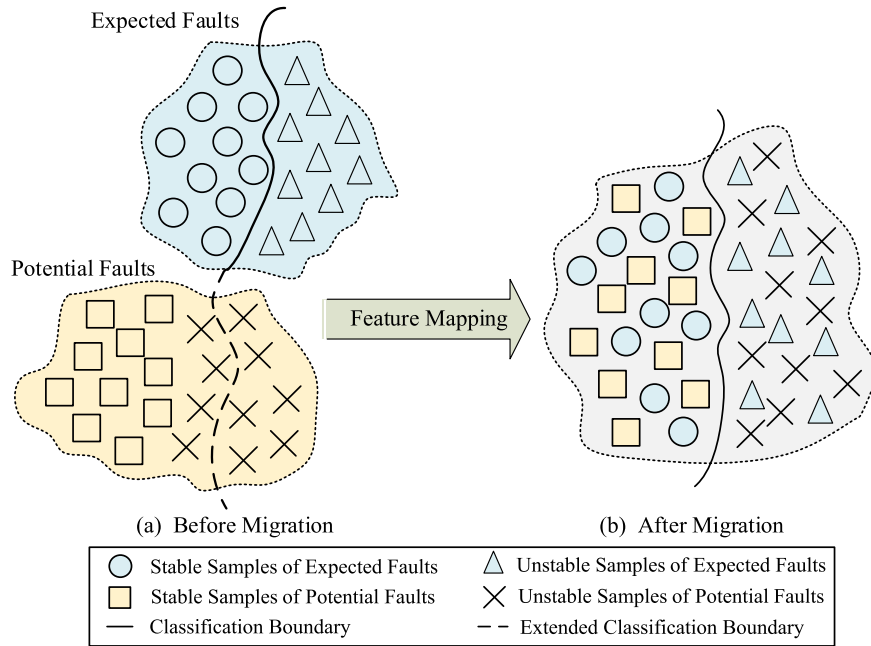


Fig. 1. : Applicable scene of transfer learning.

regular term. Therefore, the parameters of TSA model will be jointly updated with the feature distribution differences.

In short, through the transfer of knowledge, the feature distributions of expected faults and potential faults will tend to be similar. Therefore, such an updated model can be applied for the potential faults which have never been trained before.

3. Specific implementation

3.1. Module of long-short term memory

Since the fault samples are highly correlated with the temporal properties, a feature extraction module sensitive to time series is required in the model updating. RNN is a special type of neural network, which considers the correlation of data in time domain. In the hidden layer of RNN, the state variable h shows sensitivity to short-term inputs. However, it performs poorly when capturing long temporal dependencies. As an improved form of RNN, another state variable C is added to the hidden layer of LSTM. Compared with RNN, LSTM owns a stronger ability to extract features of temporal characteristics and is more suitable for dealing with a series of problems highly related to time

series. Therefore, to further explore the relationship between the temporal characteristics and transient stability, LSTM is chosen to process the original features of fault samples.

The feature extraction part of LSTM is composed of multiple repeated memory blocks. By taking outputs of the former memory block as a part of inputs of the latter one, the memory function of time series can be realized. Taking a single memory block of LSTM as an example, there exist mainly four parts, including forget gate, input gate, output gate and cell. The basic structure of LSTM is shown in Fig. 2.

Through the unit structure shown in Fig. 2, LSTM is able to simultaneously learn the information of long-term dependence and short-term dependence in time sequence, effectively solving the problems of gradient explosion and gradient disappearance. Besides LSTM can realize the memory function by setting multiple control gates, and dynamically control the weights of each gate according to the data information of each period, so as to retain the important information contained in the time series.

In summary, LSTM is selected as the model framework for feature extraction and nonlinear mapping in this paper, so as to obtain the accurate TSA results.

3.2. Time-sequence maximum mean discrepancy

For inputs of transient stability assessment, the source domain composed of expected faults is considered to be the object to be migrated. Besides, potential faults known as the target domain can be regarded as the object to be labelled.

In general, MMD is chosen as the criterion to evaluate differences in the process of transfer learning. Compared with parametric criteria, MMD evaluates a nonparametric distance and avoids the expensive distribution calculation caused by parametric criteria. However, MMD regards all the features of samples as a whole in the calculation process, ignoring the temporal properties of fault samples in power systems.

When the power system is disturbed by faults, it is obviously that the farther away from the fault clearing time, the more obvious trend of transient stability can be reflected by features (Li et al., 2020). Therefore, in this article, an improved form of MMD termed as TMMD is proposed to evaluate the differences between source domain and target

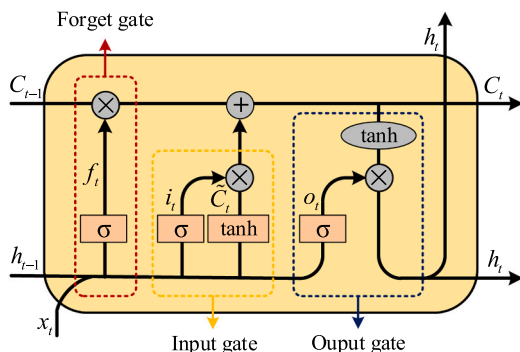


Fig. 2. Basic structure of LSTM.

domain. The essence of TMMD is to calculate the distribution differences of sample features between the source domain and target domain period by period, and then represent the importance of sample features in each period by assigning the proper weights. By doing so, TMMD can focus more on the features that are highly related to the transient stability, so as to better evaluate the distribution differences of sample features between the source domain and target domain. It is assumed that $\mathbf{p}_{1,m}$ and $\mathbf{p}_{2,m}$ are two series of probability distributions according to the features of $\mathbf{X}_{S,m}$ and $\mathbf{X}_{T,m}$ in the source domain and the target domain, TMMD can be defined as follows.

$$TMMD^2(F, \mathbf{p}_{1,m}, \mathbf{p}_{2,m}, M) = \sum_{m=1}^M \lambda_m * \sup_{f \in F}^2 \left(E_{\mathbf{p}_{1,m}} \left[f(\mathbf{X}_{S,m}) \right] - E_{\mathbf{p}_{2,m}} \left[f(\mathbf{X}_{T,m}) \right] \right) \quad (1)$$

In Eq. (1), sup function is adopted to explore the upper bound of maximum feature distribution differences between two domains, and F can be defined as a series of functions in the general reproducing kernel Hilbert space. λ represents the weight assigned to feature distribution differences within each sampling period, and M is the number of sampling periods. In addition, the sum of λ corresponding to all sampling periods shall be added up to 1.

Then, by using the reproducing property of the Hilbert space and the inner product operation property, the mean value of the expected unbiased estimation can transform the Eq. (1) into a general empirical equation that is easy to calculate. Assuming that n_S and n_T are the number of fault samples in the source domain and target domain, respectively, the empirical equation of TMMD is shown as follows.

$$TMMD_e^2(\mathbf{X}_{S,m}, \mathbf{X}_{T,m}, M) = \sum_{m=1}^M \lambda_m * \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \varphi(\mathbf{x}_{S,m}^{(i)}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \varphi(\mathbf{x}_{T,m}^{(j)}) \right\|_{\mathcal{H}}^2 \quad (2)$$

Where $\varphi: X \rightarrow \mathcal{H}$ reflects the mapping relationship of transferring original sample space to general reproducing kernel Hilbert space.

Finally, through kernel technique, the inner products between vectors are able to be solved without explicitly characterizing the mapping function $\varphi(\cdot)$, which further simplifies the calculation process of TMMD. Therefore, the nuclearized equation of TMMD can be transformed from Eq. (2) to Eq. (3) as follow.

$$TMMD_e^2(\mathbf{X}_{S,m}, \mathbf{X}_{T,m}, M) = \sum_{m=1}^M \lambda_m * \left[\frac{1}{n_S^2} \sum_{i=1}^{n_S} \sum_{j=1}^{n_S} k(\mathbf{x}_{S,m}^{(i)}, \mathbf{x}_{S,m}^{(j)}) + \frac{1}{n_T^2} \sum_{j=1}^{n_T} \sum_{i=1}^{n_T} k(\mathbf{x}_{T,m}^{(i)}, \mathbf{x}_{T,m}^{(j)}) - \frac{2}{n_S n_T} \sum_{i=1}^{n_S} \sum_{j=1}^{n_T} k(\mathbf{x}_{S,m}^{(i)}, \mathbf{x}_{T,m}^{(j)}) \right] \quad (3)$$

Where $\mathbf{x}_{S,m}^{(i)}$ stands for the feature vector of the i -th source domain sample in the m -th sampling period, $\mathbf{x}_{T,m}^{(j)}$ represents the feature vector of the j -th target domain sample in the m -th sampling period, $k(\cdot)$ stands for any kernel function in the data space. Considering that the Gaussian kernel

owns excellent generalization capability and the simple, rapid process of evaluation, it is selected as the kernel function of TMMD to participate in the calculation.

Here, we consider the exponential nonlinear weighting method for TMMD to weight the distribution differences of sample features in each period. In TMMD, the weight value λ_m of the feature distribution differences in the m -th sampling period is shown as follows.

$$\lambda_m = \frac{e^m + n}{\sum_{m=1}^M (e^m + n)} \quad (4)$$

Where n is the correction factor, which fine-tunes the distribution of weight values after normalization.

3.3. Process of online update

At the beginning of model updating, both the sample features of source domain and target domain are regarded as inputs to evaluate the distribution differences. Besides, the actual transient stability labels of expected faults are used to evaluate the cross-entropy loss of the classifier. In the process of back propagation, due to the advantage of fast calculation and the usage of momentum to alleviate the oscillation problem caused by gradient descent, the stochastic gradient descent method with momentum (Chaudhuri, 2019) is employed as the optimization tool to adjust the model parameters. The integral process of the proposed method is shown in Fig. 3.

As shown in the figure above, a domain adaptive layer is added after the module of LSTM to represent the states of fault samples in the source domain and target domain respectively. Then, the differences of feature distributions between the two domains called domain confusion loss can be evaluated accurately by TMMD.

From the forward propagation of model update, LSTM not only plays the role of feature extraction and selection, but also maps the features of fault samples to a unified high-dimensional space. Then, a back propagation (BP) neural network is taken as a simple classifier to evaluate the cross-entropy loss between actual tags and predicted tags in the source domain. When model parameters are updated by the back propagation, both the cross-entropy loss and the domain confusion loss are able to be reduced simultaneously to achieve the purpose of joint training.

Assuming that the actual label of the fault samples is y , and the prediction probability vector of transient stability labels is $\mathbf{x}$, the cross-entropy loss at the model classifier layer can be represented as follows.

$$L_{CLF} = -\log \left(\frac{\exp(\mathbf{x}[y])}{\sum_j \exp(\mathbf{x}[j])} \right) = -\mathbf{x}[y] + \log \left(\sum_j \exp(\mathbf{x}[j]) \right) \quad (5)$$

After assessing the non-parametric probability distance of feature distributions through TMMD, the value can be treated as the domain confusion loss to participate in the back propagation to update model parameters. Thus, the total loss function can be expressed as follows.

$$L_{Total} = L_{CLF} + \mu \cdot TMMD_e^2(\tilde{\mathbf{X}}_{S,m}, \tilde{\mathbf{X}}_{T,m}, M) \quad (6)$$

In Eq. (6), $\tilde{\mathbf{X}}_{S,m}$ and $\tilde{\mathbf{X}}_{T,m}$ stand for the neuronal states of the source domain samples and target domain samples in the domain adaptive layer respectively. In addition, μ is the regularization coefficient, that controls the importance of reducing feature distribution differences. The whole update process is summarized in Algorithm 1.

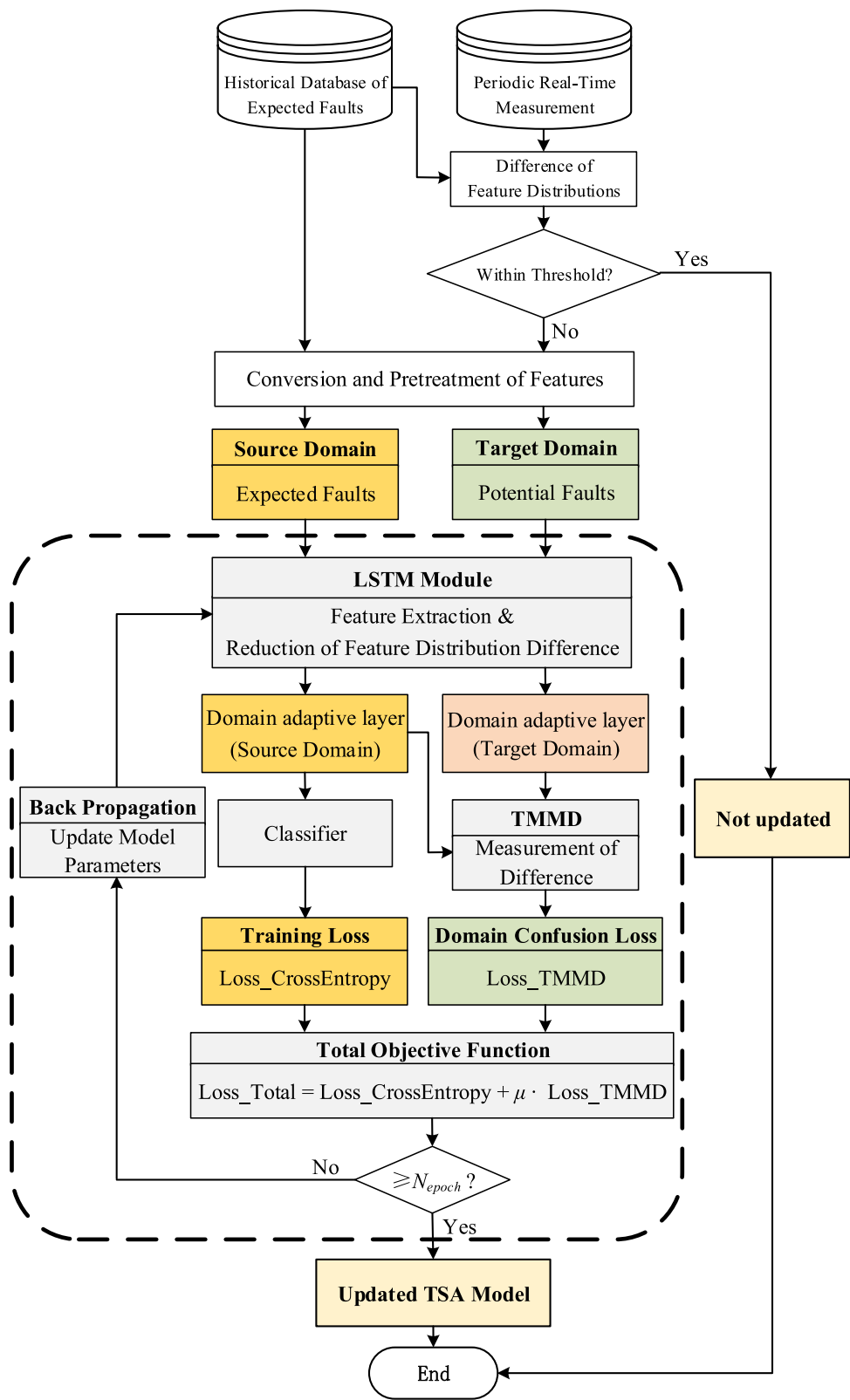


Fig. 3. The overall framework of model updating.

Algorithm 1. The online model updating algorithm

Input: Expected fault feature $X_{S,m}$ and actual label $Y_{S,m}$, potential fault feature $X_{T,m}$, regularization parameter μ .

Output: Weight-bias matrices of each memory block in LSTM module $M_1, \dots, M_k$, Weight-bias matrices of BP neural network $M_{k+1}, \dots, M_n$.

Begin

Initialize $M_1, \dots, M_k$ and $M_{k+1}, \dots, M_n$ with values of the model before updating

while the number of epochs less than or equal to N_{epoch} **do**

1. Obtain neuron states of domain adaptive layer through the forward propagation of memory blocks in LSTM module
2. Solve distribution differences between domains mapped to the unified high-dimensional feature space by Eq.(3)
3. Obtain predicted labels of expected faults in the source domain through the forward propagation of BP neural network
4. Solve cross-entropy loss between actual tags and predicted tags in the source domain by Eq.(5)
5. While updating $M_1, \dots, M_k$ and $M_{k+1}, \dots, M_n$ by stochastic gradient descent with momentum w.r.t. L_{CLF} , apply stochastic gradient descent with momentum to update $M_1, \dots, M_k$ w.r.t. $TMMD_e^2$

end while

Return the weight-bias matrices $M_1, \dots, M_k$ and $M_{k+1}, \dots, M_n$ at the end of iteration

end

4. Example analysis

In this paper, the IEEE-39 bus power system (Yang et al., 8 2018) and a realistic power system of Eastern China are selected as test systems to approve the effectiveness of the proposed method. The output results of four prediction labels are treated as the basis for analyzing the performance of TSA models. The specific definitions are shown in Table 1.

In Table 1, TP and TN represent the number of correct predictions of stable samples and unstable samples, while FN and FP are the number of false discriminations of stable samples and unstable samples. Based on the four cases of predicted labels above, the overall accuracy is adopted as the main basis to verify the superiority of the performance of the proposed method, whose definition is shown as follows.

$$\text{Test Accuracy} = \frac{TP + TN}{TP + FN + FP + TN} \quad (7)$$

4.1. Tests of IEEE-39 bus power system**4.1.1. Generation of samples**

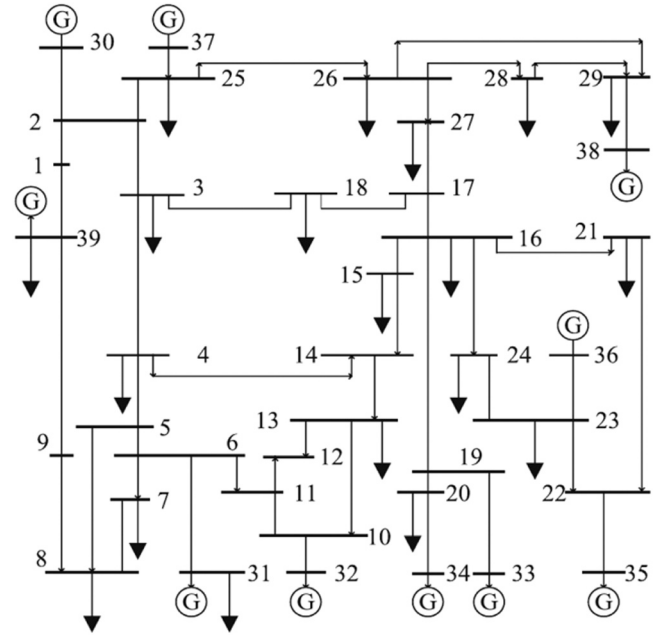
In this part, the IEEE-39 bus power system is selected as the test system. Its topological structure is presented in Fig. 4.

In the simulation, a total of 12375 fault conditions are randomly generated on PSD-BPA. Among them, 7010 stable samples and 5365 unstable samples are implied. The most serious condition of the permanent three-phase short circuit is solely adopted and a total of 5 load levels ranging from 90 % to 110 % are considered. The fault duration ranges from 100 ms to 300 ms, with 10 kinds of fault clearing moments in total. The locations of faults are set at 10 %, 50 % and 90 % of the total length of a single line. The output characteristic information includes the power angle, angular velocity deviation, reactive power and

Table 1

Predicted results of transient stability assessment.

Actual label	Predicted label	
	Stable	Unstable
Stable	TP	FN
Unstable	FP	TN

**Fig. 4.** Structure of IEEE-39 bus power system.

acceleration power of each generator in the system.

In addition, to meet the rapid response requirements of TSA, the dynamic operating information of the system in four periods is solely selected as the original features of fault samples in this paper, including the moment of fault occurrence, the moment of fault clearance, the sixth and twelfth sampling period after the fault is cleared.

4.1.2. Transfer of most lines to few lines

In the operation of power system, there always exist some lines with high reliability, which are difficult to obtain their historical faults. Besides, the faults obtained in these lines are mainly the stable samples, resulting in a serious imbalance in the sample label category. Thus, the TSA model cannot learn a complete transient stability classification boundary in the offline training process, the system instabilities caused by the faults of such lines are difficult to evaluate directly.

In this experiment, all faults on most lines are treated as the source domain. In addition, the remaining 8 lines which are difficult to obtain historical fault data are taken as the target domain. It is assumed that there only exist 12 unstable samples out of 600 fault cases from the target domain for model updating, and the remaining faults in the target domain are all treated as the test set. To verify the superiority of the method proposed in this paper, the migration effects of different transfer learning methods combined with several modules of feature extraction are compared in Table 2. Among them, Transfer Component Analysis (TCA) (Pan et al., 2011), Joint Distribution Adaptation (JDA) (Ren and Xu, 2020), MMD (Ghifary et al., 2014) and TMMD represent different methods of transfer learning respectively, while DNN, CNN and LSTM stand for different modules of feature extraction respectively.

Table 2

Transfer of lines with difficult fault acquisition.

Adopted method	Test Accuracy	
	Before migration	After migration
DNN + MMD	95.41 %	96.03 %
CNN + MMD	96.52 %	96.97 %
LSTM + TCA	96.85 %	97.02 %
LSTM + JDA	96.85 %	97.06 %
LSTM + MMD	96.85 %	97.10 %
LSTM + TMMD	96.85 %	97.83 %

It can be seen from Table 2 that all methods of transfer learning possess promoting effects on transmission lines which are difficult to obtain historical faults. Besides, the approach of this paper composed of LSTM and TMMD can achieve the best transfer effects.

Instead of training the model directly with a few faults in the target domain and expected faults in the source domain, the feature distributions of two domains can be learned and then reduced the differences through transfer learning. Even if there are a small number of faults in the target domain which are dominated by stable samples, the trained model can accurately evaluate the samples of the test set with similar feature distributions of the target domain. It is assumed that all samples of two domains are trained directly with LSTM in method 1. Besides, the proposed approach composed of LSTM and TMMD is adopted in method 2. The evaluation results of the test set are compared in Fig. 5.

As shown in Fig. 5, when only a small number of faults with stable samples in the target domain participate in the model update, the transfer learning method is superior to the direct training method. In this case, the direct training method will aggravate the evaluation tendency of stable samples, resulting in a large number of misjudgements of unstable samples. As can be seen in method 1, though the stable samples in the test set can be evaluated accurately, the unstable samples cannot be discriminated well. As for the transfer learning method, since the system instability of various faults in the target domain is sufficiently similar, the feature distributions between the source domain and the test set will also tend to be consistent while migrating the knowledge of the source domain samples to the target domain. Therefore, TSA models updated by the proposed method can evaluate both the stable samples and unstable samples well.

When enough faults are acquired and the proportion between stable samples and unstable samples is similar, due to the excellent abilities of nonlinear fitting and data mining of the model structure, the direct training method can achieve high accuracy of the test set in TSA. As for the transfer learning method, since the feature distribution differences can only be reduced to a certain extent, it is impossible to carry out accurate nonlinear fitting. Thus, the evaluation results will be slightly inferior to the direct training method.

4.1.3. Transfer of few lines to overall network

The purpose of this experiment is to explore the effect of transferring knowledge to the entire power network when all faults of a few lines are applied as the source domain.

It is assumed that all faults of the 6 randomly selected lines are taken as the source domain, then the remaining faults in the system are all

regarded as the target domain. Consistent with the experimental situation in Table 2, the migration effects of several transfer learning methods combined with feature extraction modules are compared in Table 3.

As can be seen from Table 3, under the same transfer learning method, the performance of the updated models corresponding to different feature extraction modules has been improved. Among them, LSTM performs better than ANN and CNN. Besides, on the premise of maintaining the same feature extraction module, the model updating effects of different transfer learning methods are compared in Table 3. Among them, TCA and JDA are the traditional transfer learning methods, which independently minimize the distribution differences between the source domain and target domain before updating the model parameters. Based on the idea of deep transfer learning, MMD and TMMD reduce the feature distribution differences during the updating process and retain the temporal properties of the original features, which is conducive to the feature extraction of LSTM. Thus, compared with TCA and JDA, MMD and TMMD possess better model updating effect.

Finally, the migration effects of TMMD and MMD are also compared in this experiment under the premise of maintaining other conditions. As shown in Table 3, After weighting the distribution differences of features in each period appropriately, compared with MMD, TMMD can improve the effect of knowledge transfer.

4.1.4. Transfer of single line to another single line

In this experiment, the test results before and after the migration of a single line to the other single line are compared. While proving the effectiveness of the proposed method, the relationship between the migration effect and the distance between different lines is also explored. In this section, four lines are selected as the source domain, and five lines are randomly chosen from the remaining lines as the target domain. The test results before and after migration are indicated in Fig. 6.

Where the symbol ‘bi-j’ on x-axis represents the line name between node ‘i’ and node ‘j’, and the number on y-axis stands for the test accuracy of the target domain. As shown in Fig. 6, when the source domain samples and target domain samples are under different but related fault conditions, the evaluation accuracy of the updated TSA model is improved.

In addition, according to the definition of electrical distance based on the topological structure in literature (Mao et al., 2021), the method which relied on the impedance of lines is adopted to calculate the per-unit values of equivalent electrical distances. Taking two lines of source domain transferring to two lines of target domain as examples, the equivalent distances between the lines are shown in Table 4.

For two transmission lines with similar positions, the differences of feature distributions are usually small. The corresponding instabilities of generators in the system are basically the same, which leads to the high levels of test accuracy before and after the migration. On the contrary, the feature distribution differences between two lines with a long-distance are always large, which may cause the instability differences of corresponding generators in the system. Therefore, the test accuracy before and after migration is both at low levels.

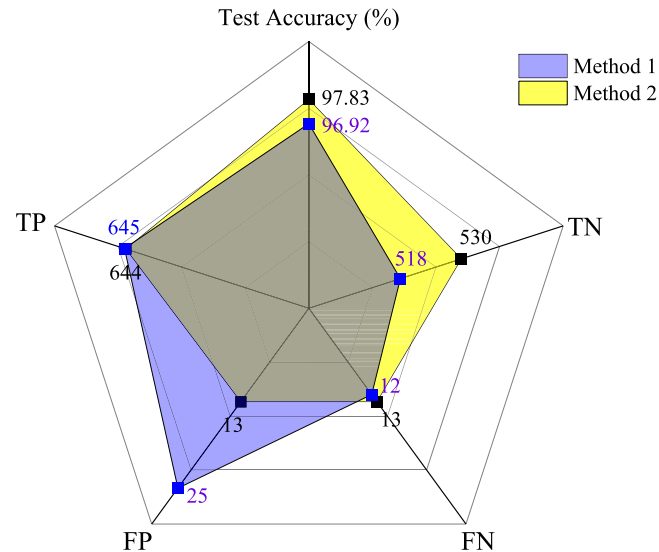


Fig. 5. Comparison of transfer learning or not.

Table 3
Transfer of few lines to the overall network.

Adopted method	Test Accuracy	
	Before migration	Before migration
DNN + MMD	90.96 %	92.84 %
CNN + MMD	92.38 %	93.75 %
LSTM + TCA	92.52 %	92.70 %
LSTM + JDA	92.52 %	92.93 %
LSTM + MMD	92.52 %	94.08 %
LSTM + TMMD	92.52 %	94.31 %

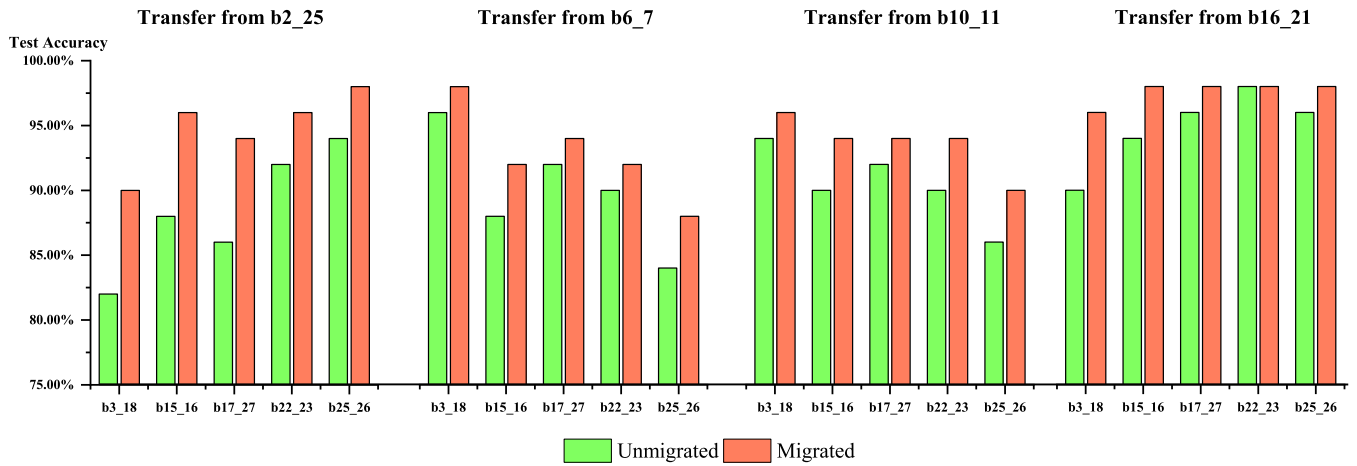


Fig. 6. Test results of single line migration.

Table 4

Equivalent distance between two lines.

Source domain	Target domain	
	b3-18	b25-26
b2-25	0.0353	0.0112
b6-7	0.0225	0.0515

4.1.5. Transfer of lines under different instability conditions

In addition, in order to verify the generality of the proposed methodology, the results of migration experiments for different instability fault cases (single-generator instability, two-generator instability, and multiple-generator instability) are also supplemented in this paper.

In the IEEE-39 bus power system, the transient instability phenomena caused by line faults can be roughly categorised into three cases: single machine instability at bus 38, two machine instability at bus 37 and 38, and multiple machine instability at bus 30–38. In this experiment, 1 line with a total of 1125 expected faults is selected from each instability case to constitute the source domain, and 1 line with a total of 375 fault samples is randomly selected from the remaining lines as the target domain. Assuming that only 25 target domain samples have actual transient stability labels, the remaining potential faults are used as a test set to measure the performance of the TSA model.

As can be seen from the results in Table 5, when the failure instability mode of the power system changes, the proposed model can also effectively improve the evaluation accuracy.

4.1.6. Effects of different TMMD weights on migration

As for the method proposed in this paper, the feature distribution differences evaluated by TMMD are treated as a regular term of the total loss function to update the model. Thus, the weight offered to the regular term will inevitably affect the migration effect. Taking the overall migration as an example, 6 lines are randomly chosen as the source

Test Accuracy (%)

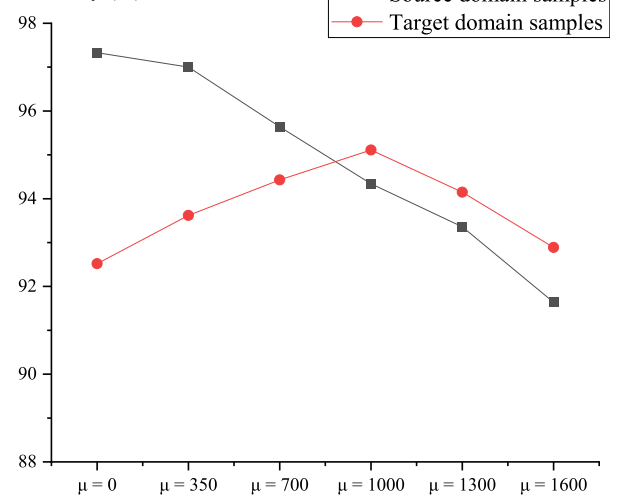


Fig. 7. Influence of weight on domain confusion loss.

domain, and all the remaining lines are regarded as the target domain. Gradually changing the weight of domain confusion loss, its influences on the transfer effect are shown in Fig. 7, where μ is the weight of domain confusion loss in Eq. (6).

As indicated in Fig. 7, when the weight of the domain confusion loss increases, the training accuracy of source domain decreases continuously, while the test accuracy of target domain increases first and then decreases. At first, when μ increases from 0, before reaching a certain threshold, reducing the differences of feature distributions at the cost of a small amount of source domain training accuracy can effectively improve the performance of TSA models. When μ reaches the threshold, the updated TSA model achieves a minimized state of the relative trade-off between the cross-entropy loss in the source domain and the distribution differences between source domain and target domain. At this point, the updated TSA model is in the optimal evaluation state. When μ exceeds the threshold and continues to increase, overemphasizing the feature distribution differences and ignoring the prior knowledge based on expected faults will result in negative effects, which may ruin the evaluation ability of the TSA model.

In short, two factors are considered to select the weight value of μ . On the one hand, the magnitude order of the domain confusion loss shall be approximately equal to that of the cross-entropy loss. On the other hand,

Table 5

Test results for different instability conditions.

Faulty instability condition	Test Accuracy	
	Before migration	After migration
Single-generator	93.71 %	95.43 %
Two-generator	95.14 %	96.86 %
Multiple-generator	94.28 %	96.28 %

Table 6
Effects of weight in different sampling periods.

	Before Migration	MMD	TMMD
Period 1	-	0.25	0.03
Period 2	-	0.25	0.09
Period 3	-	0.25	0.24
Period 4	-	0.25	0.64
Test Accuracy	93.19 %	94.45 %	94.81 %

the specific value of μ is determined by the evaluation results of the validation set.

4.1.7. Influence of weights in different sampling periods

For the faults applied in TSA, the relationship between the features from different sampling periods and transient stability are different. In this part, the effects of different weights offered to each sampling period are discussed. Taking the overall migration as an example, 8 lines are randomly selected as the source domain, and the samples corresponding to the remaining lines are all treated as the target domain.

In this experiment, MMD and the proposed TMMD are respectively applied to evaluate the differences of feature distributions between fault samples in the source domain and the target domain. Taking a total of four sampling periods into account, on the basis of applying LSTM as the model framework, the transfer effects under different measurement methods are shown in Table 6.

In Table 5, the data from row 2 to row 5 represent the weights assigned by different measurement methods to the feature distribution differences in each period. As can be seen from Table 5, the model updating effect of the proposed TMMD is superior to that of MMD method. This phenomenon indicates that the proposed TMMD method is more consistent with the transient stability trajectory of power system after disturbance. Compared with MMD, TMMD can effectively improve the effect of transfer learning to update TSA models.

4.2. Tests of realistic regional power system

In order to indicate the practicability and scalability of the proposed method, a realistic regional power system of Eastern China is employed as the test system. Specific details of the system are shown in Table 7.

In realistic regional power system, transient stability of the system is mainly determined by EHV transmission lines and UHV transmission lines. Therefore, in 597 transmission lines with voltage levels of 50 kV and 100 kV, all faults included in 120 lines are regarded as the source domain. Besides, two series of 15 non-repeating lines are chosen from the remaining transmission lines as the objects to be transferred. The detailed breakdown configurations in the realistic regional power system are shown in Table 8.

Based on the detailed configurations mentioned above, a total of 7200 fault conditions are generated. To achieve the migration from source domain to target domain, 30 % of the samples in target domain are selected to participate in the updating process, and 70 % of the remaining faults are adopted as the basis to evaluate the effects of transfer learning. The test results transferring to two series of the target domains are shown in Fig. 8.

In this experiment, two series of completely different lines are chosen as target domains respectively to verify the transfer effects of different regions in the system. As depicted in Fig. 8, in the realistic regional power system, both MMD and TMMD can contribute to the TSA of potential faults. In addition, TMMD with higher weights assigned to the

Table 7
Details of realistic regional system.

Components	Details
Bus	6040 Buses (10–1000 kV)
Transmission Line	5599 Lines (10–1000 kV)
Generator	705 Generators (10–1080 MW)
Maximum Capacity	305140 MW
Operation Frequency	50 Hz

Table 8
Details of fault conditions in practical system.

Indexes	Details
Source Domain	120 Lines (500 kV and 1000 kV)
Target Domain	15 Lines (500 kV and 1000 kV)
Fault Type	Permanent Three-Phase Short Circuit
Load Level	100 %
Fault Location	10 %, 50 % and 90 % of Each Lines
Fault Duration	5 Cycles to 20 Cycles

later sampling periods can highlight the part of features more closely related to transient stability, which performs better than MMD. In short, both the test results of two scenarios above indicate that the proposed approach of this paper is also applicable in the large-scale power systems.

5. Conclusions

In this paper, by utilizing LSTM module to extract the features of fault samples, a transfer learning method is proposed for the updating of the TSA model. Based on temporal properties of the transient stability, TMMD is designed to quantify the feature distribution differences in time series between the expected faults and potential faults. Moreover, the quantified differences are appended to the total loss function to update the parameters of the model.

To illustrate the performance of the proposed approach, two test cases of power systems are taken into account, including the IEEE-39 bus power system and a realistic regional power system. All the evaluation results reveal that the proposed method for model updating can achieve a superior transfer capability. Meanwhile, TMMD with higher weights assigned to the features of later periods can better evaluate the differences of feature distributions between expected faults and potential faults.

In addition, this paper also studies the characteristics of feature distribution differences between two lines in different regions. The experimental results suggest that the distribution differences of sample features are closely related to the electrical distance between lines.

In the proposed approach, TMMD is adopted as the criterion to evaluate the feature distribution differences between expected faults and potential faults. Considering the peculiarities of fault samples for TSA, there may be other evaluation criteria which can further improve the capability of migration.

The proposed transfer learning updating method can effectively update the model to adapt to potential faults when the power system operating condition changes greatly. However, the proposed updating method still has some limitations. First of all, in order to achieve sufficient evaluation performance of the updated model, it is necessary to ensure that there are certain potential scenario samples. However, when the number of potential scene samples is large enough, the update effect

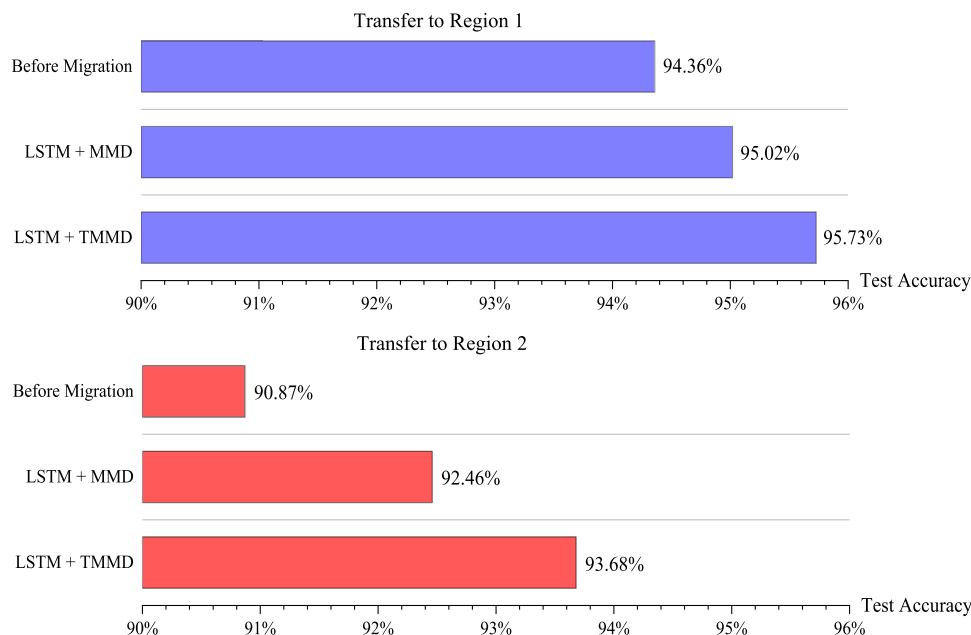


Fig. 8. Effects of MMD and TMMD in practical system.

of fine tuning will be superior to that of transfer learning. Due to these shortcomings, it is necessary to explore other new fault feature distribution difference evaluation criteria in the future to further improve the model migration ability.

CRedit authorship contribution statement

Yongbin Li: Writing – original draft, Validation. **Jian Li:** Validation, Formal analysis. **Mingkai Yang:** Writing – original draft, Software, Project administration, Methodology, Conceptualization. **Yihua Fu:** Writing – review & editing. **Xi Fang:** Writing – review & editing. **Guo-qiang Lu:** Validation, Formal analysis. **Huaiyuan Wang:** Funding acquisition, Conceptualization.

Declaration of Competing Interest

We declare that all authors have no known competing financial interests or personal relationships that could have appeared to influence the review of the manuscript entitled “Transfer Learning-Based Updating Method of Transient Stability Assessment Model”.

Data availability

The datasets used or analyzed during the current study are available from the corresponding author on reasonable request.

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